Maintenance Schedule

Diesel engine
12V4000P91
16V4000P91
Application group 4C

M050777/03E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
12V4000P91	145 kW/cyl.	4C, short time operation
16V4000P91	145 kW/cyl.	4C, short time operation

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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1.3 Load profile

Load profile

Power	110%	100%	75%	50%
Corresponding operating time	1%	29%	40%	30%

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1.4 Maintenance schedule matrix

0 to 9,000 Operating hours

Task	Limit	Operating hours [h]																		
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000
W0534	1 m																			
W1053	1 a																			
W1024	1 a																			
W1008	2 a																			
W0500	-	X																		
W0501	-	X																		
W0502	-	X																		
W0503	-	X																		
W0504	-	X																		
W0505	-	X																		
W0506	-	X																		
W0507	-	X																		
W0508	-	X																		
W0525	-	X																		
W1001	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1047	6 m			X		X		X		X		X		X		X		X		X
W1463	1 a			X		X		X		X		X		X		X		X		X
W1002	2 a			X		X		X		X		X		X		X		X		X
W1209	1 a				X			X			X			X			X			X
W1046	2 a							X						X						X
W1005	3 a							X						X						X
W1006	9 a													X						
W1011	9 a													X						
W1043	9 a																			X
W1684	9 a																			X
W2000	9 a																			X
W2001	9 a																			X
W2002	9 a																			X
W2003	9 a																			X
W2004	9 a																			X
W2006	9 a																			X
W2007	9 a																			X
W2008	9 a																			X
W2010	9 a																			X
W2011	9 a																			X
W2070	9 a																			X
W2062	9 a																			X
W2110	9 a																			X
W1026	5 a																			X
W1462	5 a																			X
W1250	6 a																			X
W1251	6 a																			X

w = weeks
m = months
a = years

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Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000											
W1058	9 a																			X											
W1051	9 a																			X											
W1134	9 a																			X											
W1038	18 a																			X											
W3000	18 a																			X											
W3001	18 a																			X											
W3002	18 a																			X											
W3003	18 a																			X											
W3004	18 a																			X											
W3005	18 a																			X											
W3007	18 a																			X											
W3012	18 a																			X											
W3013	18 a																			X											
W3021	18 a																			X											
W3100	18 a																			X											

w = weeks
m = months
a = years

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1.5 Maintenance tasks

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Task
QL1	-	1 m	Engine operation	Test run at not below 1/3 load and at least until steady-state temperature is reached.	W0534
QL1	-	1 a	Lubrication points	Apply lubricant at lubrication points.	W1053
QL1	-	1 a	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).	W1008
QL1	Daily	-	Engine operation	Check engine oil level.	W0500
				Carry out visual inspection of engine for general condition and leaks.	W0501
				Inspect intercooler drain system.	W0502
				Inspect service indicator of air filter.	W0503
				Check the inspection bores of the HP fuel pump.	W0504
				Check relief bores of coolant pump(s).	W0505
				Check for abnormal running noises, exhaust gas color and vibration.	W0506
				Drain off water and contamination from fuel prefilter.	W0507
				Check differential pressure gauge of fuel prefilter.	W0508
				Inspect battery-charging generator for contamination and clean if necessary.	W0525
QL1	500	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.	W1001
QL1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.	W1009
QL1	1000	6 m	Oil indicator filter	Check and clean oil indicator filter.	W1047
QL1	1000	1 a	Engine mounts	Carry out visual inspection of engine mounts for general condition.	W1463
QL1	1000	2 a	Valve gear	Check valve clearance, adjust if necessary.	W1002
QL1	1500	1 a	Battery charging alternator	Remove battery-charging generator and clean thoroughly with compressed air.	W1209
QL1	3000	2 a	Crankcase breathers	Fit new filters or filter inserts.	W1046
QL1	3000	3 a	Air filters	Fit new air filters.	W1005
QL1	6000	9 a	Fuel injectors	Replace fuel injectors.	W1006
QL1	6000	9 a	Combustion chambers	Inspect cylinder chambers using endoscope.	W1011
QL3	9000	9 a	Battery charging alternator	Overhaul battery-charging generator.	W1043
QL3	9000	9 a	Starter	Replace copper beryllium pinion, after 300 starts, at the latest.	W1684
QL3	9000	9 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.	W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.	W2001
				Clean air ducting.	W2002
				Clean intercooler and inspect for leakage.	W2003
				Fit new high-pressure fuel sensor.	W2004
				Check engine coolant thermostat and fit new thermal actuator.	W2006
				Check charge-air coolant thermostat and fit new thermal actuator.	W2007
				Overhaul engine coolant preheating system.	W2008
				Overhaul starter.	W2010
				Inspect vibration damper.	W2011
				Overhaul charge-air coolant pump.	W2070
				Fit new seals/sealing materials for all disassembled components.	W2062
				Overhaul engine coolant pump.	W2110

w = weeks
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Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Task
QL3	9000	5 a	Engine mounts	Check buffer clearance of mounts, check proper seating of securing screws.	W1026
				Check for cracks. Measure cracks if necessary. Measure height of rubber elements.	W1462
QL3	9000	6 a	Rubber sleeves	Replace all rubber sleeves.	W1250
QL3	9000	6 a	Hose lines	Replace all hose lines.	W1251
QL3	9000	9 a	HP fuel pump	Fit new high-pressure fuel pump.	W1058
QL3	9000	9 a	Fuel delivery pump	Fit new fuel delivery pump.	W1051
QL3	9000	9 a	Cylinder heads	Overhaul cylinder heads.	W1134
QL3	9000	18 a	Turbochargers	Overhaul turbochargers.	W1038
QL4	9000	18 a	Extended component maintenance	Completely disassemble the engine.Inspect engine components as per assembly instructions and repair or fit new components as required.	W3000
				Replace all elastomeric parts and seals with new ones.	W3001
				Fit new piston rings.	W3002
				Fit new conrod bearings.	W3003
				Fit new crankshaft bearings.	W3004
				Fit new cylinder liners.	W3005
				Fit new high-pressure fuel pump.	W3007
				Fit new actuator cylinders for exhaust flaps.	W3012
				Fit new exhaust flap bearings.	W3013
				Fit new pressure relief valve in high-pressure fuel system.	W3021
				Fit new rubber elements of engine mounts.	W3100
w = weeks m = months a = years					

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